

Exceptional points, Puiseux series, and Symmetry-allowed Hessian forms

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Why Non-Hermitian (NH) Systems?

Not a new topic ...

- ▶ Dissipative systems ➤ Experiments
- ▶ Open, non-equilibrium systems
 - interaction with surrounding environment
- ▶ Effective description of systems with finite lifetime τ
 - one-body quasiparticle Hamiltonian is non-Hermitian as $\text{Im}[\Sigma] \sim \tau^{-1}$
- ▶ Photonic systems with gain or loss

Need basic theory involving “Complex Hamiltonians”

NH Systems: Specific Examples

- ▶ **Optical model for scattering of nucleons** ➡ **single particle wavefunction satisfies Schrödinger eqn with NH optical potential**

[F. Coester & H. Kümmel, *Nuclear Physics* (1958)]

- ▶ **Decaying states in time-dependent perturbation theory arising as bound states of 2 particles** ➡ **NH potential**

[M. M. Sternheim & J. F. Walker, *PRB* (1972)]

- ▶ **Classical mechanical, electrical, optical systems with dissipation**

Complex Matrix: Exceptional Points

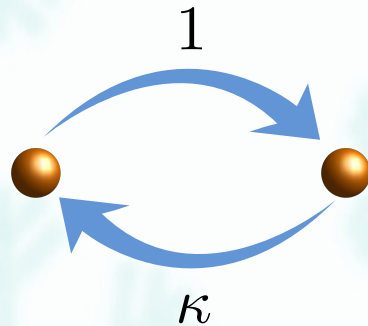
Singular points of Complex Matrix → **Matrix becomes non-diagonalizable**

→ **Two or more eigenvalues coincide**

+ corresponding eigenvectors coalesce into one

A minimal example

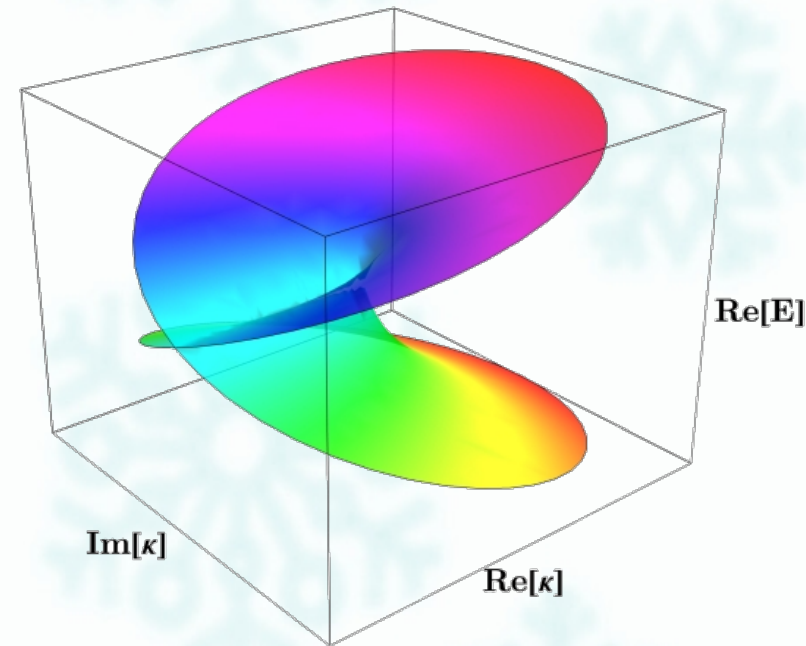
$$H = \begin{pmatrix} 0 & \kappa \\ 1 & 0 \end{pmatrix}$$



Eigenvalues $E = \pm\sqrt{\kappa}$

Non-orthogonal right and left eigenvectors:

$$|\Psi_{R,\pm}\rangle = \begin{pmatrix} \pm\sqrt{\kappa} \\ 1 \end{pmatrix}, \quad \langle\Psi_{L,\pm}| = (1 \quad \pm\sqrt{\kappa})$$

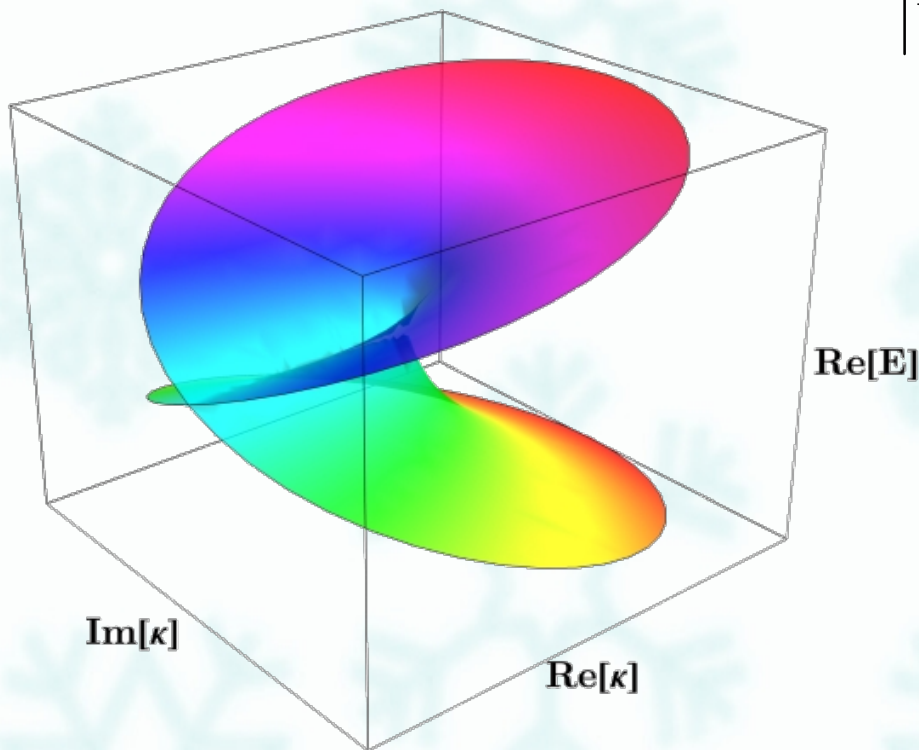


Exceptional Points: Minimal Example

Exceptional Point (EP) ☛

$\kappa = 0 \Rightarrow E = 0$ (doubly degenerate)
Eigenvectors also coalesce into one:

$$|\Psi_{R,\pm}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \langle \Psi_{L,\pm}| = (1 \quad 0)$$

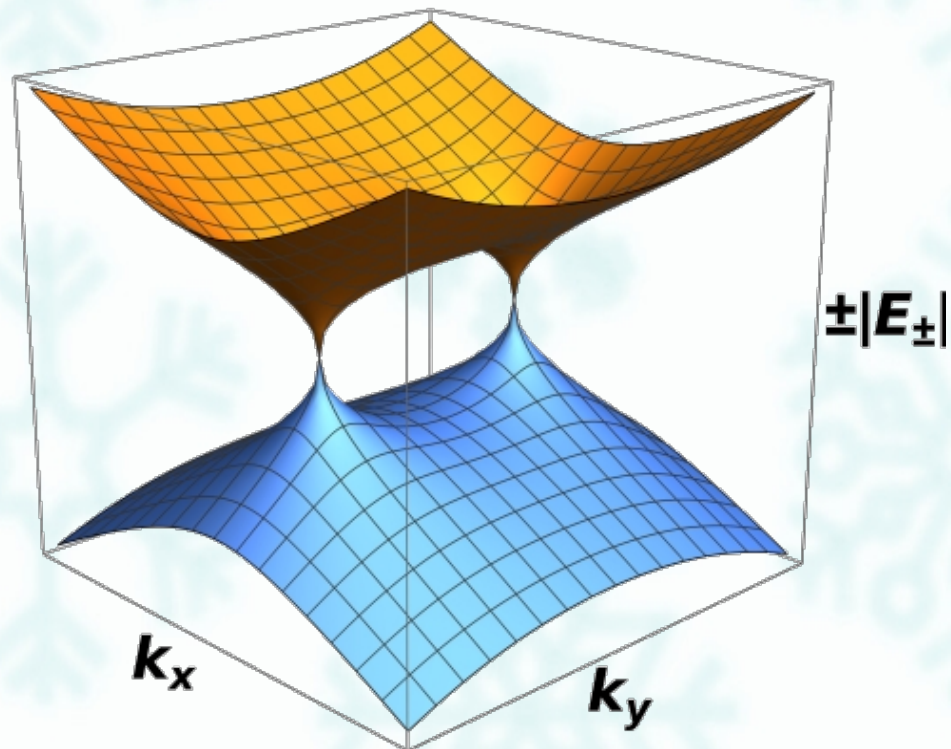


- ▶ Incomplete set of linearly-independent eigenvectors at EP
☛ H cannot be diagonalized
- ▶ Winding of κ twice about EP yields winding of E once
- ▶ EP is a branch point

When can we expect EPs to occur?

What are their consequences?

Exceptional Nodal Phases



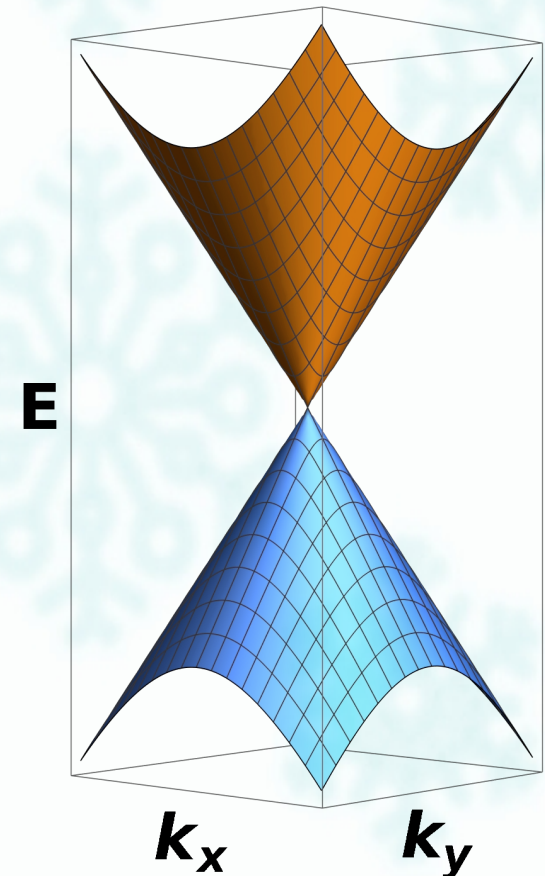
Stable Band Crossing: Topological Phase

$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma} = d_1(\mathbf{k}) \sigma_x + d_2(\mathbf{k}) \sigma_y + d_3(\mathbf{k}) \sigma_z$$

$$E = \pm \sqrt{d_1^2(\mathbf{k}) + d_2^2(\mathbf{k}) + d_3^2(\mathbf{k})}$$

Hermitian system $\Rightarrow d_1, d_2, d_3$ real

- ▶ Need 3 equations to get a gap closing point $\Rightarrow d_1 = d_2 = d_3 = 0$
- ▶ Fine-tuned in $d=2$
- ▶ Stable / generic in $d=3$ as 3 equations can be solved for 3 momenta $\{k_x, k_y, k_z\}$
 - $\Rightarrow$ simplest example: **Weyl node**



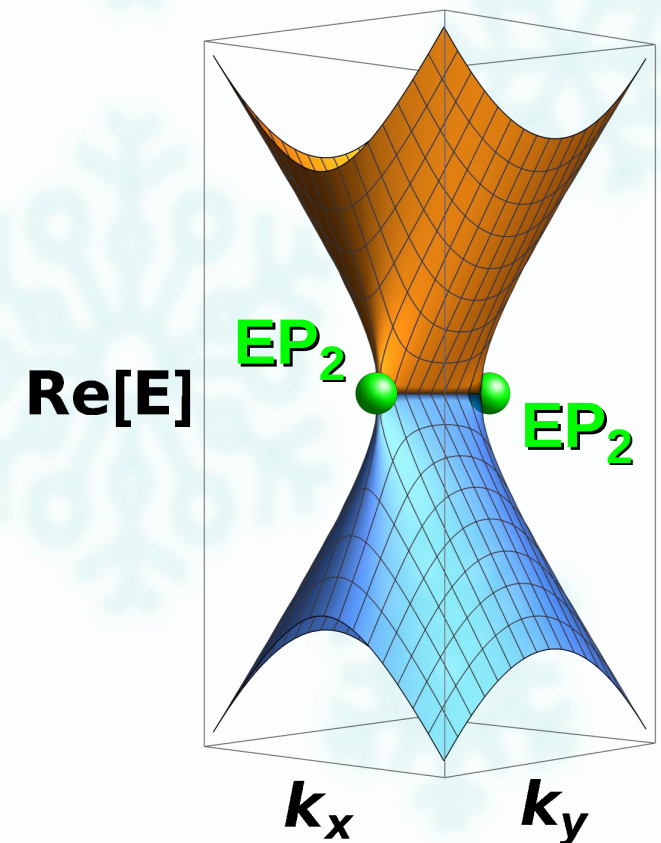
Non-Hermitian (NH) Topology

$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad \mathbf{d}(\mathbf{k}) = \mathbf{d}_R(\mathbf{k}) + i \mathbf{d}_I(\mathbf{k})$$

$$E = \pm \sqrt{\mathbf{d}_R^2(\mathbf{k}) - \mathbf{d}_I^2(\mathbf{k}) + 2i \mathbf{d}_R(\mathbf{k}) \cdot \mathbf{d}_I(\mathbf{k})}$$

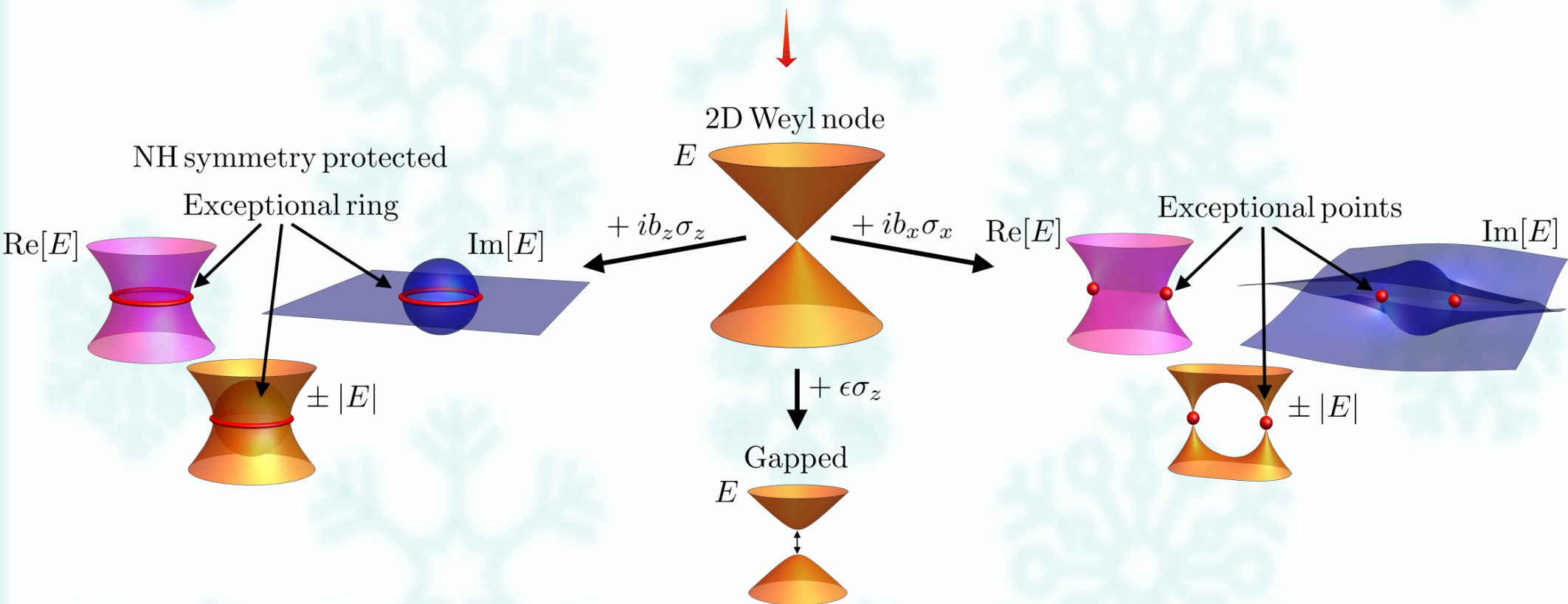
Non-Hermitian system

- ▶ Need 2 equations to get a gap closing point
- ▶ Stable/generic in $d = 2$ as 2 equations can be solved for 2 momenta $\{k_x, k_y\}$
- ▶ Degeneracies appear as EPs
 - come in pairs



Splitting Weyl Nodes

$$H = k_x \sigma_x + k_y \sigma_y$$



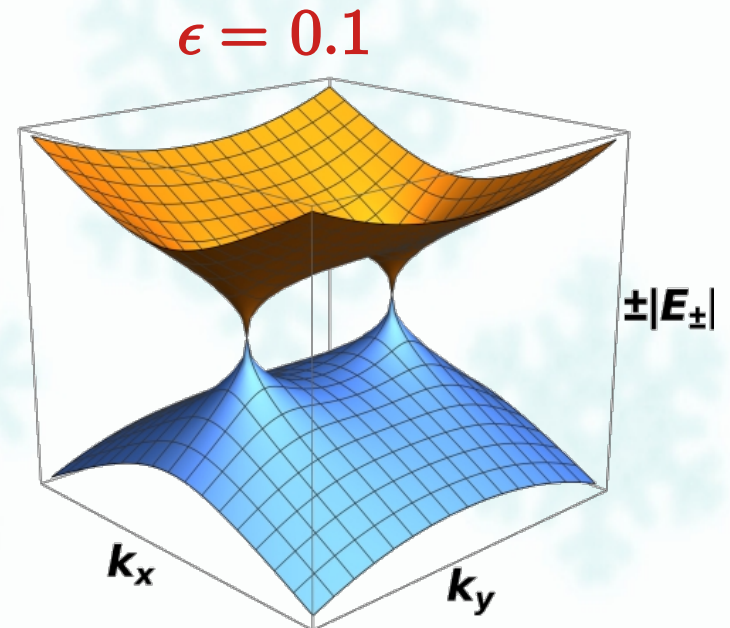
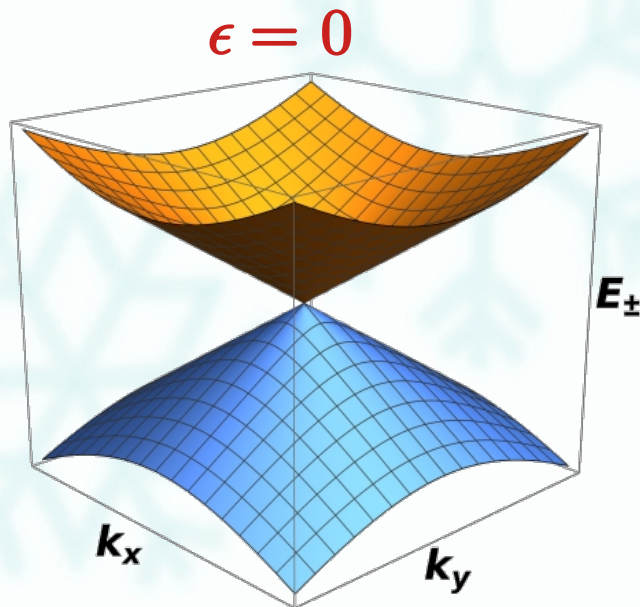
[Bergholtz et al (2020)]

Spectral Features

- ▶ **EPs are non-analytical** ➡ “square roots of Weyl points”
 - ➡ $E \propto \sqrt{|\mathbf{k} - \mathbf{k}_{EP}|}$
- ▶ **EPs come in pairs + generic in 2d**
 - ➡ much more abundant than in Hermitian case

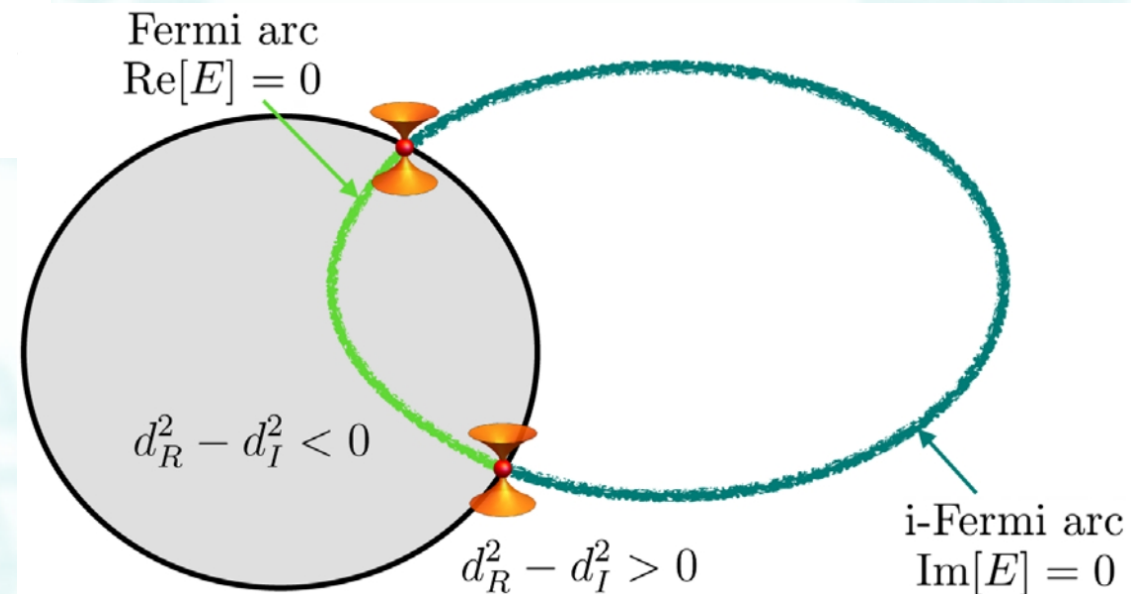
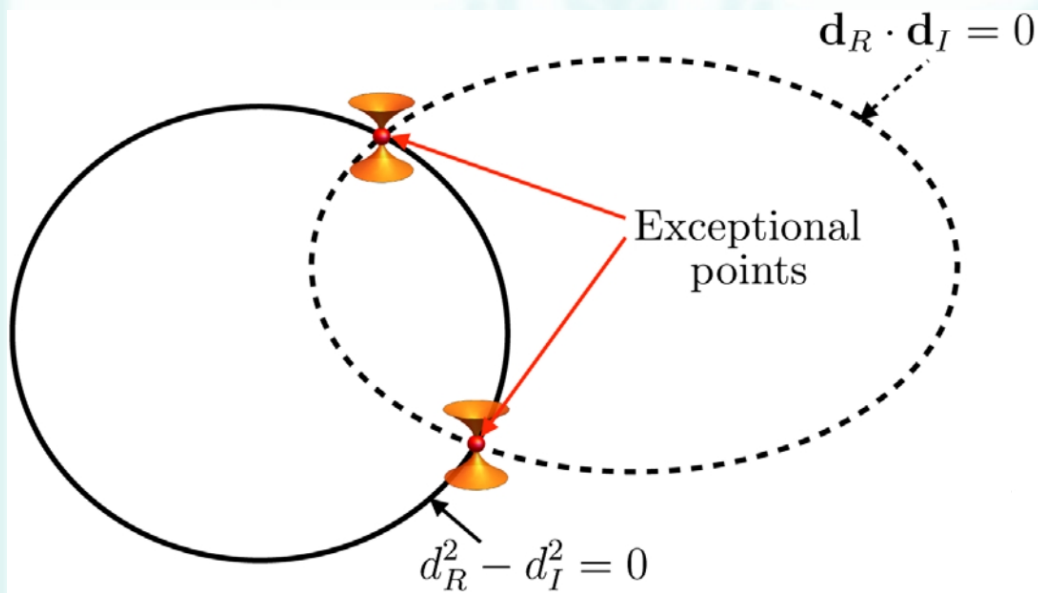
$$H(\mathbf{k}) = k_x \sigma_x + k_y \sigma_y + i\epsilon \sigma_x \Rightarrow E_{\pm} = \pm \sqrt{k_x^2 + k_y^2 - \epsilon^2 + 2i\epsilon k_x}$$

$$\text{EP at } \{k_x = 0, k_y = \pm\epsilon\}$$



Stable NH Phases

$$E = \pm \sqrt{\mathbf{d}_R^2(\mathbf{k}) - \mathbf{d}_I^2(\mathbf{k}) + 2i \mathbf{d}_R(\mathbf{k}) \cdot \mathbf{d}_I(\mathbf{k})}$$

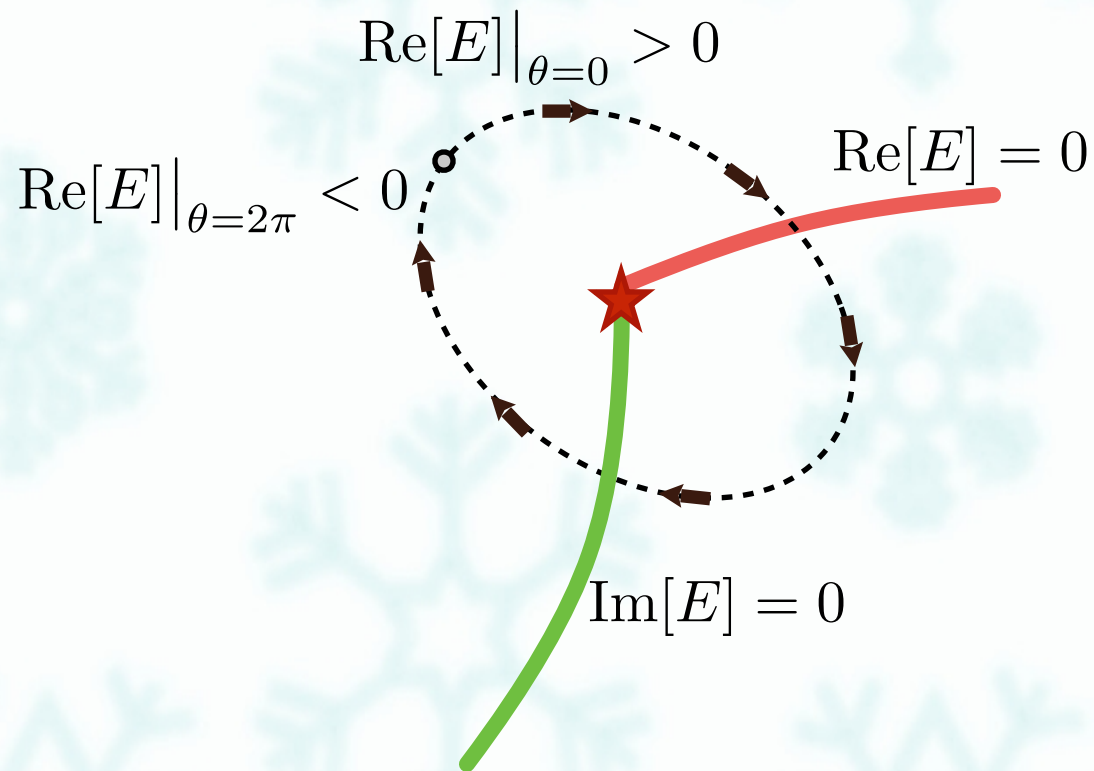


[Bergholtz et al (2020)]

Generic (d-1)-dimensional open nodal surfaces / arcs $\leftrightarrow$ Stable degeneracies

Around an EP

Move around an EP_2 , tracking an eigenstate



$$|\Psi_{R,\pm}\rangle = \begin{pmatrix} \pm r e^{i\theta/2} \\ 1 \end{pmatrix}$$

We end up in the other eigenstate after one closed orbit

↪ 4π periodicity

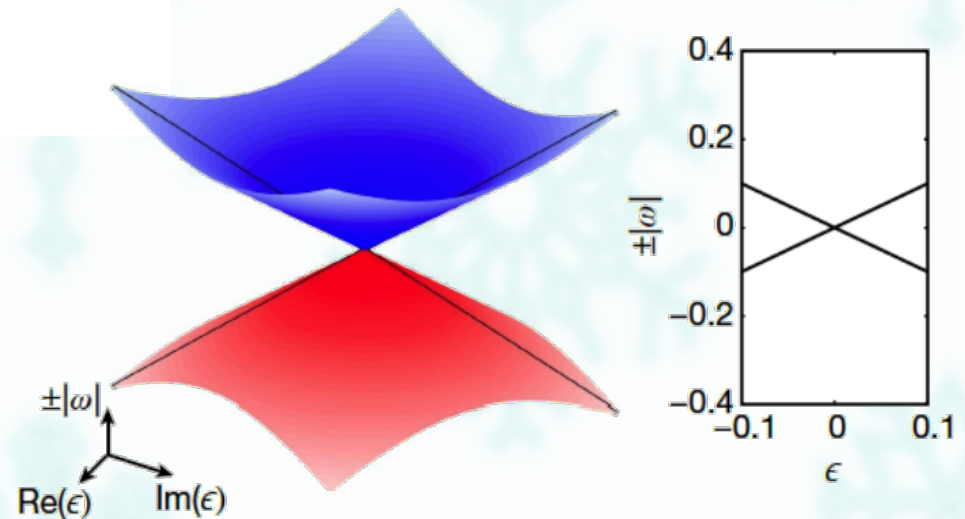
Enhanced Sensitivity at EPs

Exceptional points enhance sensing in an optical microcavity

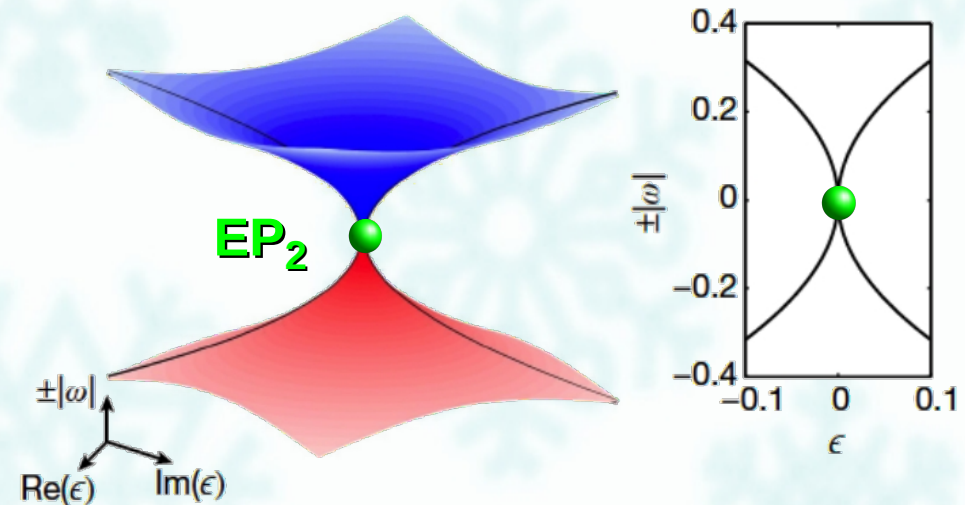
Weijian Chen, Şahin Kaya Özdemir, Guangming Zhao, Jan Wiersig & Lan Yang ✉

Nature **548**, 192–196(2017)

Energy splitting $\propto \epsilon$

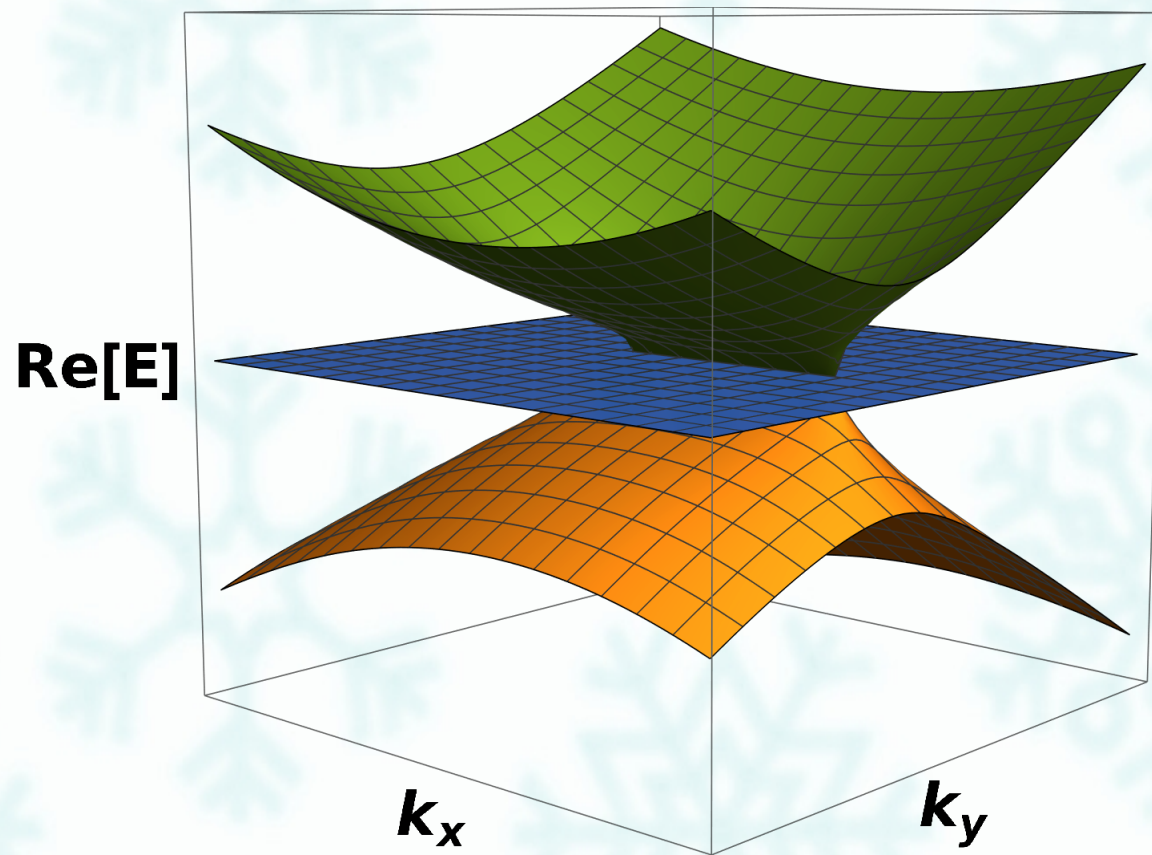


Energy splitting $\propto \epsilon^{1/2}$



Non-Hermitian sensing devices operating at Exceptional points / rings
☞ **enhanced sensitivity**

Higher-Order Exceptional Points



Higher-Order EPs: Number of Constraints

- ▶ Any complex square matrix is triangularizable (Schur's lemma)
- ▶ Eigenvalues are the diagonal entries of this upper or lower triangular form
- ▶ Hence, a generic $N \times N$ non-Hermitian complex matrix can be brought into the form:

$$\begin{pmatrix} E_1 & 0 & \dots & 0 \\ \kappa_{21} & E_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \kappa_{N1} & \kappa_{N2} & \dots & E_N \end{pmatrix}$$

$$\begin{pmatrix} E & 0 & \dots & 0 \\ \kappa_{21} & E & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \kappa_{N1} & \kappa_{N2} & \dots & E \end{pmatrix} \quad \text{has an EP of order } N$$

Need $(N-1)$ complex constraint equations to get N^{th} order EP

Higher-Order EPs: My Work

- ▶ N^{th} order EP = $\text{EP}_N \iff N$ Eigenvalues + N Eigenvectors coalesce
- ▶ Need $N \times N$ complex matrix
- ▶ Need $(2N-2)$ dimensions for stable EP_N

Simplest Case $\iff \text{EP}_3$

- ▶ Need $(2 \times 3) - 2 = 4$ dimensions for stable EP_3
- ▶ Physical systems have $d \leq 3 \iff d = 4$ not possible
- ▶ Generic NH symmetries / constraints $\iff$ stable EP_3 in $d = 2$

Parity (P) symmetry

$$H = -p H p^\dagger$$
$$p p^\dagger = \mathbb{1}$$

Parity-Time (PT) symmetry

$$H = (p k) H^* (p k)^\dagger$$
$$k k^\dagger = \pm \mathbb{1}$$

3-Band Model With P-Symmetry

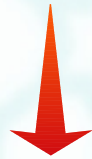
$$H_P(\mathbf{k}) = -PH_P(\mathbf{k})P^{-1} \text{ with } P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow H_P = \begin{pmatrix} 0 & b & 0 \\ d & 0 & f \\ 0 & h & 0 \end{pmatrix}$$

where $b, d, f, h \in \mathbb{C}$

$$E = 0, \pm \sqrt{bd + fh}$$

$$\Rightarrow \text{EP}_3 \text{ at } b = -\frac{fh}{d}$$

$$\text{with } |\psi_R\rangle = \left(-\frac{f}{d}, 0, 1\right)^T, \langle\psi_L| = \left(-\frac{h}{b}, 0, 1\right)$$



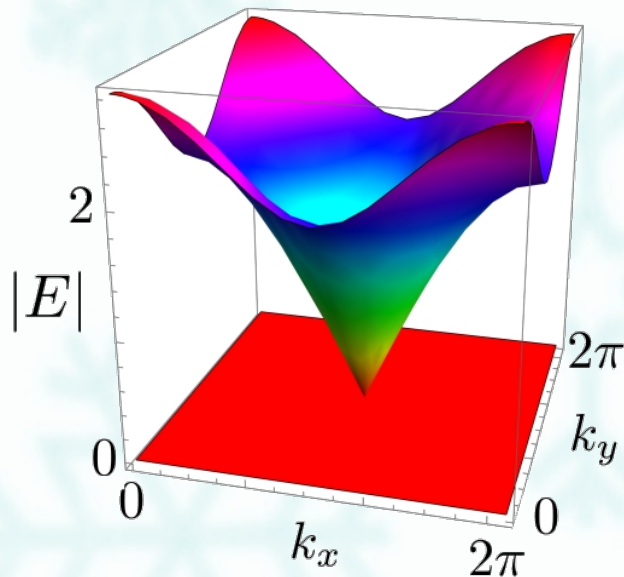
2 real constraints  **Generic EP₃ in 2d**

3-Band Model With P-Symmetry

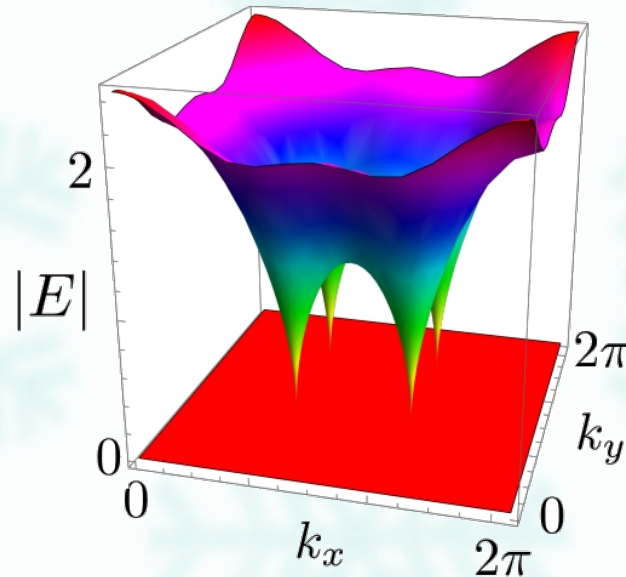
Physical System ➡ NH Lieb lattice model

$$H_{\text{Lieb}} = \begin{pmatrix} 0 & 1 + e^{i k_x} - i \epsilon & 0 \\ 1 + e^{-i k_x} - i \epsilon & 0 & 1 + e^{i k_y} + i \epsilon \\ 0 & 1 + e^{-i k_y} + i \epsilon & 0 \end{pmatrix}$$

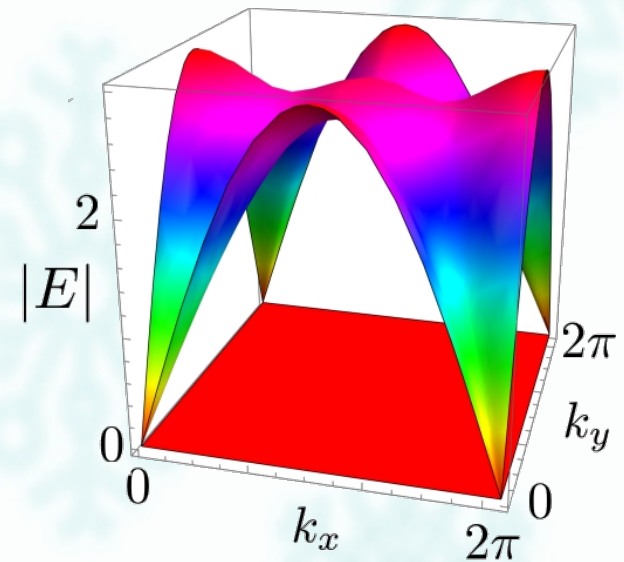
$\epsilon = 0$



$\epsilon = 1$

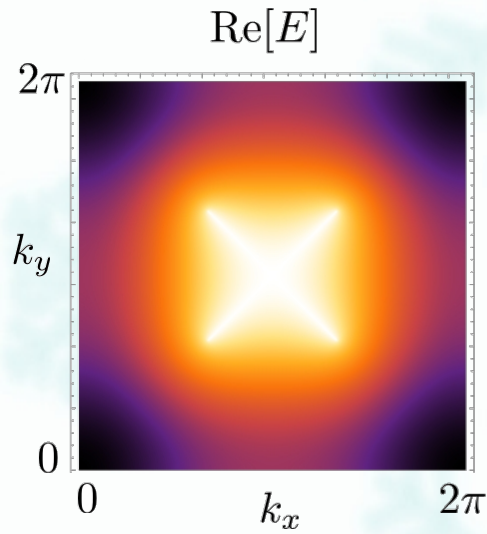


$\epsilon = 2$

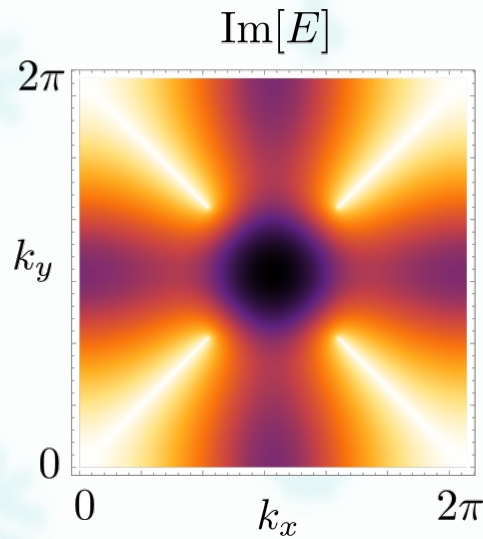


3-Band Model With P-Symmetry

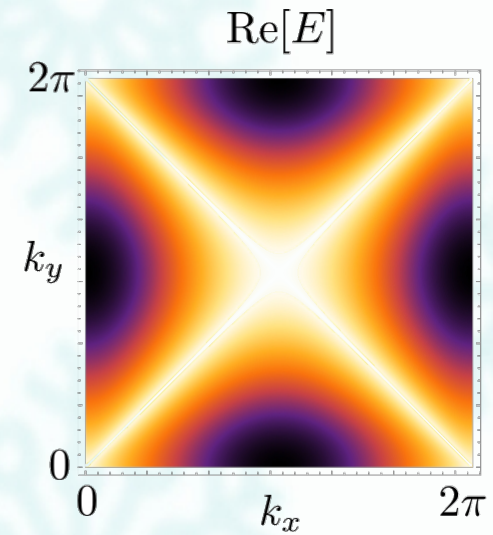
Fermi arcs 🖐



$\epsilon = 1$



$\epsilon = 1$



$\epsilon = 2$

[IM & E. J. Bergholtz, *PRL* (2021)]

3-Band Model With PT-Symmetry

$$H_{\text{PT}} = (P K) H_{\text{PT}}^* (P K)^{-1} \quad \text{with } K = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & i \end{pmatrix}$$

$$\Rightarrow H_{\text{PT}} = \begin{pmatrix} \text{real}_1 & i \text{real}_2 & \text{real}_3 (1 - i) \\ i \text{real}_4 & \text{real}_5 & \text{real}_6 (1 + i) \\ \text{real}_7 (1 + i) & \text{real}_8 (1 - i) & \text{real}_9 \end{pmatrix}$$

Eigenvalues ➡ **Cardano's method:**

$$E_1 = \alpha_+ + \alpha_-, \quad E_2 = \omega \alpha_+ + \omega^* \alpha_-, \quad E_3 = \omega^* \alpha_+ + \omega \alpha_-$$

$$\alpha_{\pm} \equiv \sqrt[3]{q \pm \sqrt{p^3 + q^2}}, \quad \omega = \frac{-1 + \sqrt{3}i}{2}, \quad \text{with } p, q \in \mathbb{R}$$

$$\Rightarrow \text{EP}_3 \text{ at } p = q = 0$$

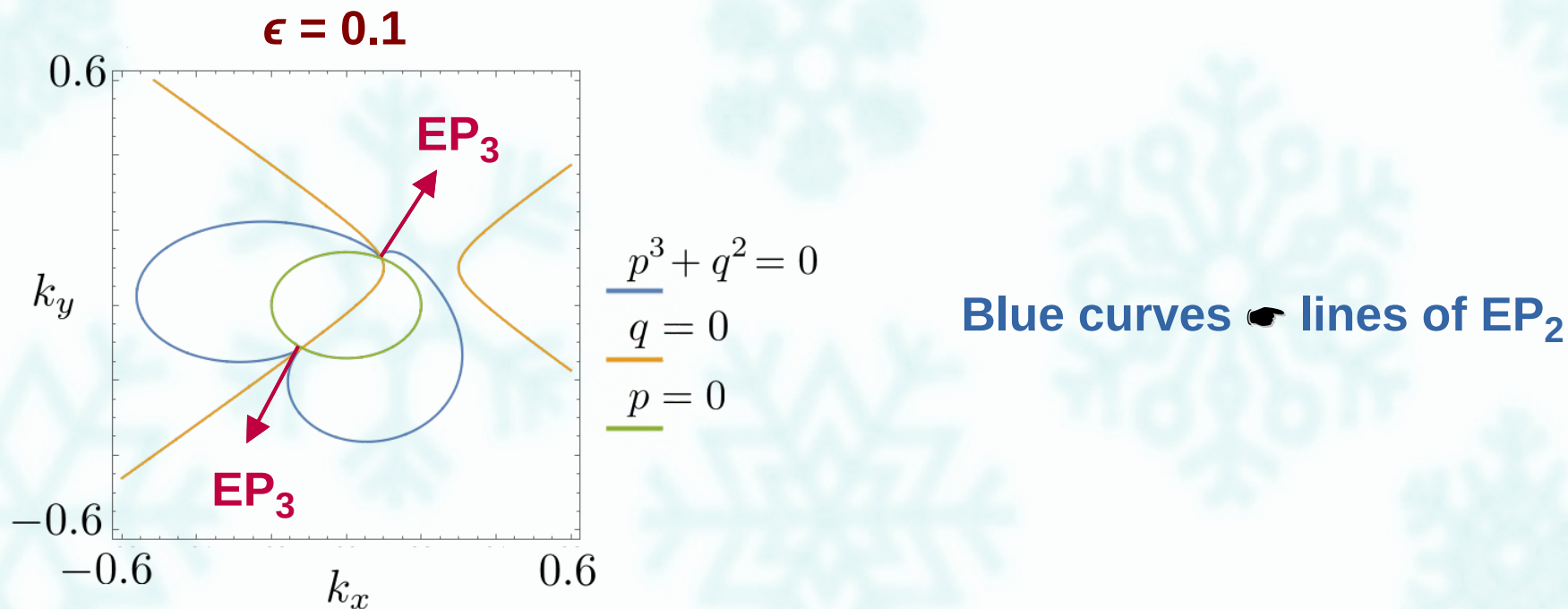
 **2 real constraints** ➡ **Generic EP₃ in 2d**

EP₃ in 2d: Case 2

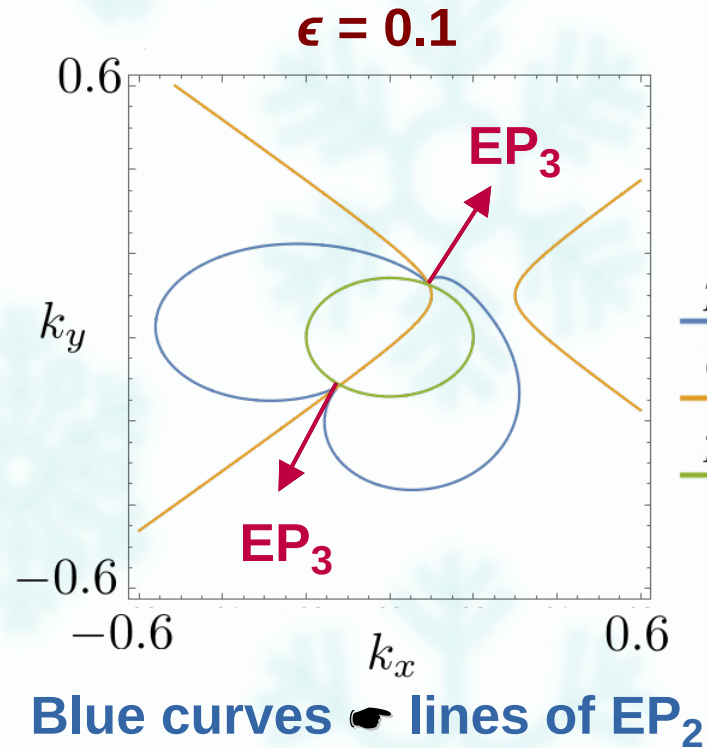
$$H_{PT} = (p \ k) H_{PT}^* (p \ k)^\dagger$$

Linearised Dirac-like NH Hamiltonian obeying PT-symmetry

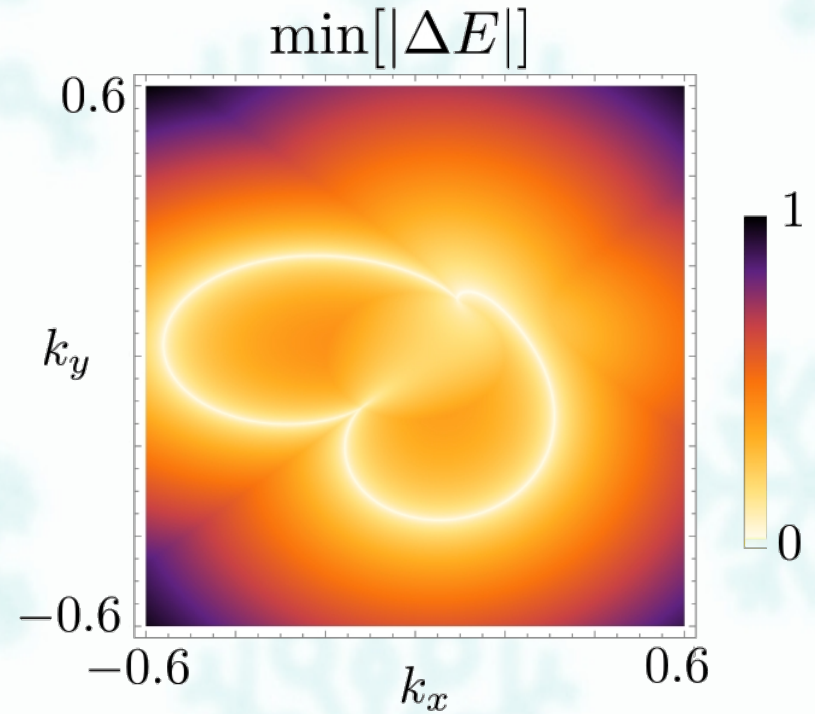
$$H_{PT} = \begin{pmatrix} -\epsilon & k_x + \epsilon & (1+i)\epsilon \\ k_x - \epsilon & 0 & (1+i)(k_y - \epsilon) \\ 2i\epsilon & (1-i)(k_y + \epsilon) & \epsilon \end{pmatrix}$$



3-Band Model with PT-Symmetry

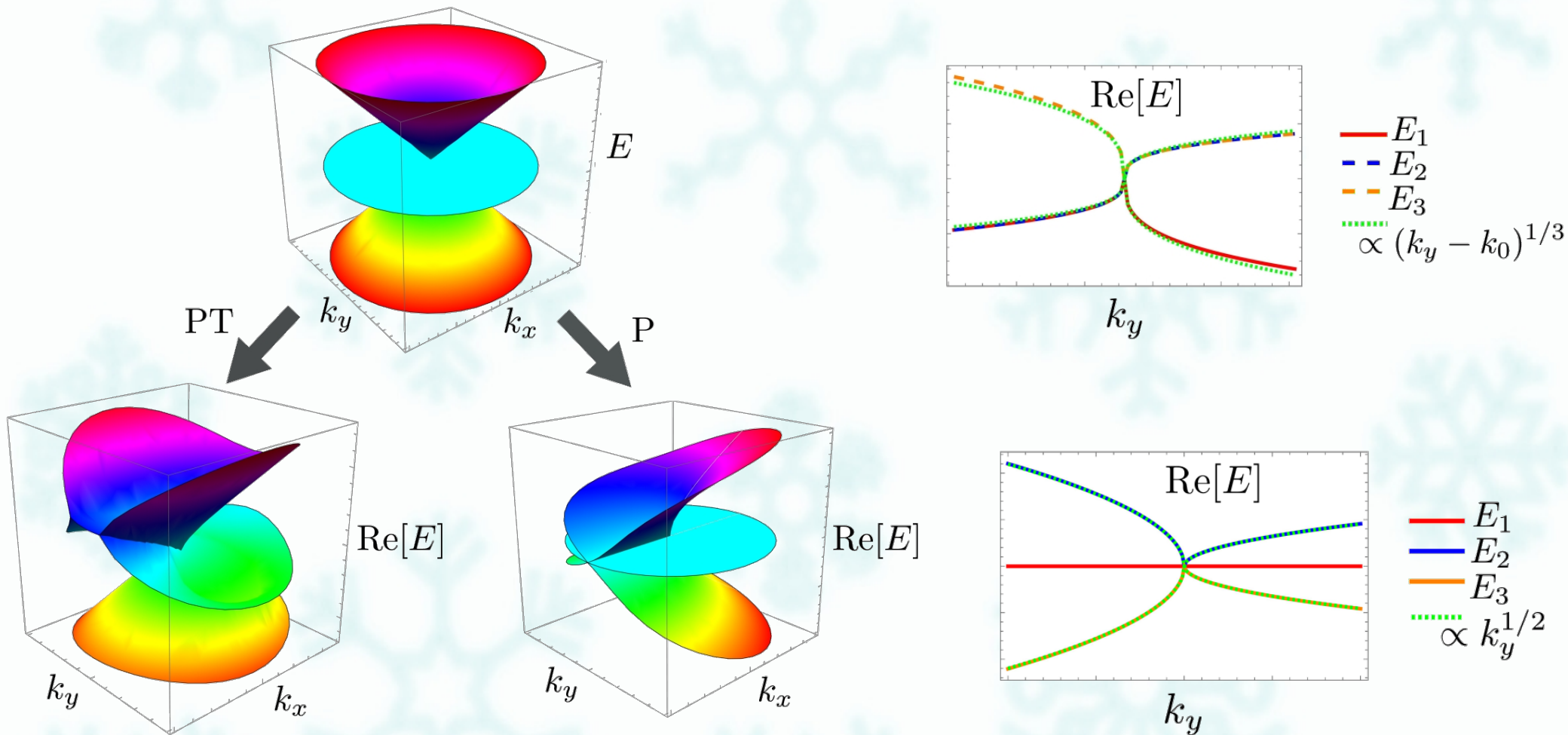


$$\begin{aligned} \text{---} p^3 + q^2 &= 0 \\ \text{---} q &= 0 \\ \text{---} p &= 0 \end{aligned}$$



[IM & E. J. Bergholtz, *PRL* (2021)]

EP₃'s: Contrasting Singular Behaviour



Different symmetries stabilize branch points of different orders

Energy splitting $\propto k^{1/3}$ for PT

[Puiseux series]

$\propto k^{1/2}$ for P

[IM & E. J. Bergholtz, *Phys. Rev. Lett.* (2021)]

EP₃: Experimental Realization

Experimental simulation of symmetry-protected higher-order exceptional points with single photons

KUNKUN WANG , LEI XIAO, HAIQING LIN, WEI YI , EMIL J. BERGHOLTZ , AND PENG XUE  [Authors Info & Affiliations](#)

SCIENCE ADVANCES • 23 Aug 2023 • Vol 9, Issue 34 • DOI: 10.1126/sciadv.adi0732

- ▶ Experimentally simulated 2d NH band structures using single-photon interferometry
- ▶ Observed stable EP₃s by tuning only 2 real parameters

PHYSICAL REVIEW LETTERS **130**, 263601 (2023)

Editors' Suggestion

Featured in Physics

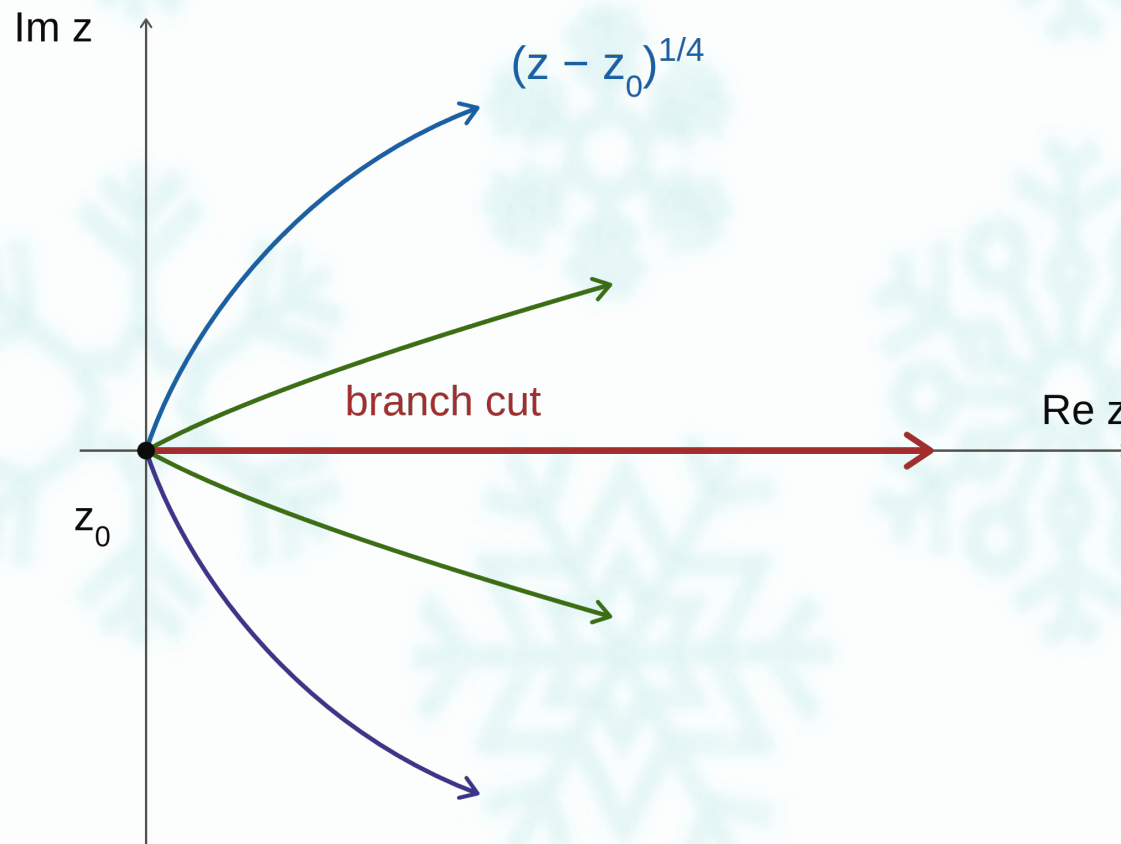
Observation of Exceptional Points in Thermal Atomic Ensembles

Chao Liang,¹ Yuanjiang Tang ,¹ An-Ning Xu,¹ and Yong-Chun Liu ^{1,2,*}

¹State Key Laboratory of Low-Dimensional Quantum Physics, Department of Physics,
Tsinghua University, Beijing 100084, China

²Frontier Science Center for Quantum Information, Beijing 100084, China

Puiseux Series $\rightarrow$ Power series in deviation from EP, admitting fractional exponents



Hessenberg Matrices

Upper- ℓ Hessenberg matrix $B_{n,\ell}$ of dim. $n \times n$

• $(\ell-1)$ nonzero sub-diagonals below principal diagonal

$$B_{n,\ell} = \begin{pmatrix} m_{1,1} & m_{1,2} & m_{1,3} & \cdots & \cdots & \cdots & m_{1,n} \\ m_{2,1} & m_{2,2} & m_{2,3} & \cdots & \cdots & \cdots & m_{2,n} \\ \vdots & \vdots & \vdots & \ddots & & & \vdots \\ m_{\ell,1} & m_{\ell,2} & m_{\ell,3} & \cdots & \cdots & \cdots & m_{\ell,n} \\ 0 & m_{\ell+1,2} & m_{\ell+1,3} & \cdots & \cdots & \cdots & m_{\ell+1,n} \\ 0 & 0 & m_{\ell+2,3} & \cdots & \cdots & \cdots & m_{\ell+2,n} \\ \vdots & \vdots & \ddots & \ddots & \ddots & & \vdots \\ 0 & 0 & \cdots & 0 & m_{n,n-\ell+1} & \cdots & m_{n,n} \end{pmatrix}$$

Hessenberg Matrices

$$n = 2$$

$$B_{2,1} = \begin{pmatrix} m_{1,1} & m_{1,2} \\ 0 & m_{2,2} \end{pmatrix} \quad B_{2,2} = \begin{pmatrix} m_{1,1} & m_{1,2} \\ m_{2,1} & m_{2,2} \end{pmatrix}$$

$$n = 3$$

$$B_{3,1} = \begin{pmatrix} m_{1,1} & m_{1,2} & m_{1,3} \\ 0 & m_{2,2} & m_{2,3} \\ 0 & 0 & m_{3,3} \end{pmatrix} \quad B_{3,2} = \begin{pmatrix} m_{1,1} & m_{1,2} & m_{1,3} \\ m_{2,1} & m_{2,2} & m_{2,3} \\ 0 & m_{3,2} & m_{3,3} \end{pmatrix}$$

$$B_{3,3} = \begin{pmatrix} m_{1,1} & m_{1,2} & m_{1,3} \\ m_{2,1} & m_{2,2} & m_{2,3} \\ m_{3,1} & m_{3,2} & m_{3,3} \end{pmatrix}$$

Perturbation around EP_n

$$J_n(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ 0 & 0 & \lambda & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}_{n \times n}$$

Perturbed matrix around EP_n

$$J_n(\lambda) + \epsilon B_{n,\ell}$$

How do degenerate eigenvalues of J_n split?

Perturbation around EP_n

Nature of eigenvalues under all possible perturbed matrices
↪ characterised by Puiseux series

- ▶ Prepare all possible sets of $\{n_i\}$ with $\sum_i n_i = n$
- ▶ For each i we have a ring of size n_i , defined by

$$\lambda_{i,p_{n_i}} = \lambda + \alpha_{i,1} \omega_{s_i}^{p_{n_i}} \epsilon^{\frac{1}{n_i}} + \alpha_{i,2} \omega_{n_i}^{2p_{n_i}} \epsilon^{\frac{2}{n_i}} + \dots$$
$$\omega_{n_i} = e^{\frac{2\pi i}{n_i}} \text{ for } p_{n_i} \in [1, n_i]$$

[Y. Ma & A. Edelman, *Linear Algebra ...* (1998)]

Puiseux Series about EP_2 : $n = 2$

2 sets of rings: $\{n_1 = 1, n_2 = 1\}$ & $\{n_1 = 2\}$

Ring index $i \in \{1, 2\}$ & sizes $\in \{n_1 = 1, n_2 = 1\}$

$$\lambda_{1,1} = \lambda + \alpha_{1,1} \epsilon + \alpha_{1,2} \epsilon^2 + \cdots ,$$

$$\lambda_{2,1} = \lambda + \alpha_{2,1} \epsilon + \alpha_{2,2} \epsilon^2 + \cdots$$

Ring index $i \in \{1\}$ & size $\in \{n_1 = 2\}$

$$\lambda_{1,1} = \lambda - \alpha_{1,1} \epsilon^{\frac{1}{2}} + \alpha_{1,2} \epsilon + \cdots , \quad \lambda_{1,2} = \lambda + \alpha_{1,1} \epsilon + \alpha_{1,2} \epsilon^2 + \cdots$$

Puiseux Series about EP_3 : $n = 3$

3 sets of rings: $\{n_1 = 1, n_2 = 1, n_3 = 1\}$, $\{n_1 = 2, n_2 = 1\}$, $\{n_1 = 3\}$

Ring index $i \in \{1, 2, 3\}$ & sizes $\in \{n_1 = 1, n_2 = 1, n_3 = 1\}$

$$\lambda_{1,1} = \lambda + \alpha_{1,1} \epsilon + \alpha_{1,2} \epsilon^2 + \cdots ,$$

$$\lambda_{2,1} = \lambda + \alpha_{2,1} \epsilon + \alpha_{2,2} \epsilon^2 + \cdots ,$$

$$\lambda_{3,1} = \lambda + \alpha_{3,1} \epsilon + \alpha_{3,2} \epsilon^2 + \cdots$$

Ring index $i \in \{1, 2\}$ & sizes $\in \{n_1 = 2, n_2 = 1\}$

$$\lambda_{1,1} = \lambda - \alpha_{1,1} \epsilon^{\frac{1}{2}} + \alpha_{1,2} \epsilon + \cdots , \quad \lambda_{1,2} = \lambda + \alpha_{1,1} \epsilon^{\frac{1}{2}} + \alpha_{1,2} \epsilon + \cdots ,$$

$$\lambda_{2,1} = \lambda + \alpha_{2,1} \epsilon + \alpha_{2,2} \epsilon^2 + \cdots$$

Ring index $i \in \{1\}$ & size $\in \{n_1 = 3\}$

$$\lambda_{1,1} = \lambda + \alpha_{1,1} \omega_3 \epsilon^{\frac{1}{3}} + \alpha_{1,2} \omega_3^2 \epsilon^{\frac{2}{3}} + \alpha_{1,3} \omega_3^3 \epsilon + \cdots ,$$

$$\lambda_{1,2} = \lambda + \alpha_{1,1} \omega_3^2 \epsilon^{\frac{1}{3}} + \alpha_{1,2} \omega_3^4 \epsilon^{\frac{2}{3}} + \alpha_{1,3} \omega_3^6 \epsilon + \cdots ,$$

$$\lambda_{1,3} = \lambda + \alpha_{1,1} \omega_3^3 \epsilon^{\frac{1}{3}} + \alpha_{1,2} \omega_3^6 \epsilon^{\frac{2}{3}} + \alpha_{1,3} \omega_3^9 \epsilon + \cdots$$

Perturbation around EP_3 : P-symmetric

$$\mathcal{H}_3^{\text{P}} = \begin{bmatrix} 0 & b & c \\ d & 0 & 0 \\ g & 0 & 0 \end{bmatrix}, \quad P = \text{diag}\{-1, 1, 1\}, \quad P \mathcal{H}_3^{\text{P}} P^{-1} = -\mathcal{H}_3^{\text{P}}$$

$$S^{-1} \mathcal{H}_{\text{EP}}^{\text{P}} S = \mathcal{J}_3(0), \quad \text{with } S = \begin{bmatrix} 0 & \frac{1}{g} & 0 \\ \frac{d}{g} & 0 & -\frac{d}{c g^2} \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and } \mathcal{H}_{\text{EP}}^{\text{P}} = \mathcal{H}_3^{\text{P}} \Big|_{b = -\frac{c g}{d}}$$

$$\tilde{B}_{3,2}^{\text{P}} = \begin{bmatrix} 0 & a_1 & a_2 \\ a_3 & 0 & 0 \\ a_4 & 0 & 0 \end{bmatrix}, \quad S^{-1} \tilde{B}_{3,2}^{\text{P}} S = \begin{bmatrix} 0 & \frac{a_4}{g} & 0 \\ a_1 d + a_2 g & 0 & -\frac{a_1 d}{c g} \\ 0 & c \left(a_4 - \frac{a_3 g}{d} \right) & 0 \end{bmatrix}$$

P-symmetry-preserving perturbation ➡ **upper-2 Hessenberg matrix**

Perturbation around EP_3 : P-symmetric

$$S^{-1} \mathcal{H}_{\text{EP}}^{\text{P}} S = \mathcal{J}_3(0)$$

P-symmetry-preserving perturbation $\rightarrow$ **upper-2 Hessenberg matrix**

Split eigenvalues

$$\left\{ 0, \pm \sqrt{c \left(a_4 - \frac{a_3 g}{d} \right) + a_1 (a_3 \epsilon + d) + a_2 (a_4 \epsilon + g) \epsilon^{1/2}} \right\}$$

Perturbation around EP_3 : PT-symmetric

$$H_3^{\text{PT}} = \begin{bmatrix} \text{real}_1 & i \text{real}_2 & \text{real}_3 (1 - i) \\ i \text{real}_4 & \text{real}_5 & \text{real}_6 (1 + i) \\ \text{real}_7 (1 + i) & \text{real}_8 (1 - i) & \text{real}_9 \end{bmatrix},$$

$$P = \text{diag}\{1, -1, 1\}, \quad K = \text{diag}\{1, 1, i\}$$

$$H_3^{\text{PT}} = (P K) (H_3^{\text{PT}})^* (P K)^{-1}$$

PT-symmetry-preserving perturbation ➡ upper-3 Hessenberg matrix

➡ eigenvalues $\sim \epsilon^{1/3}$

Leading-Order Singularity

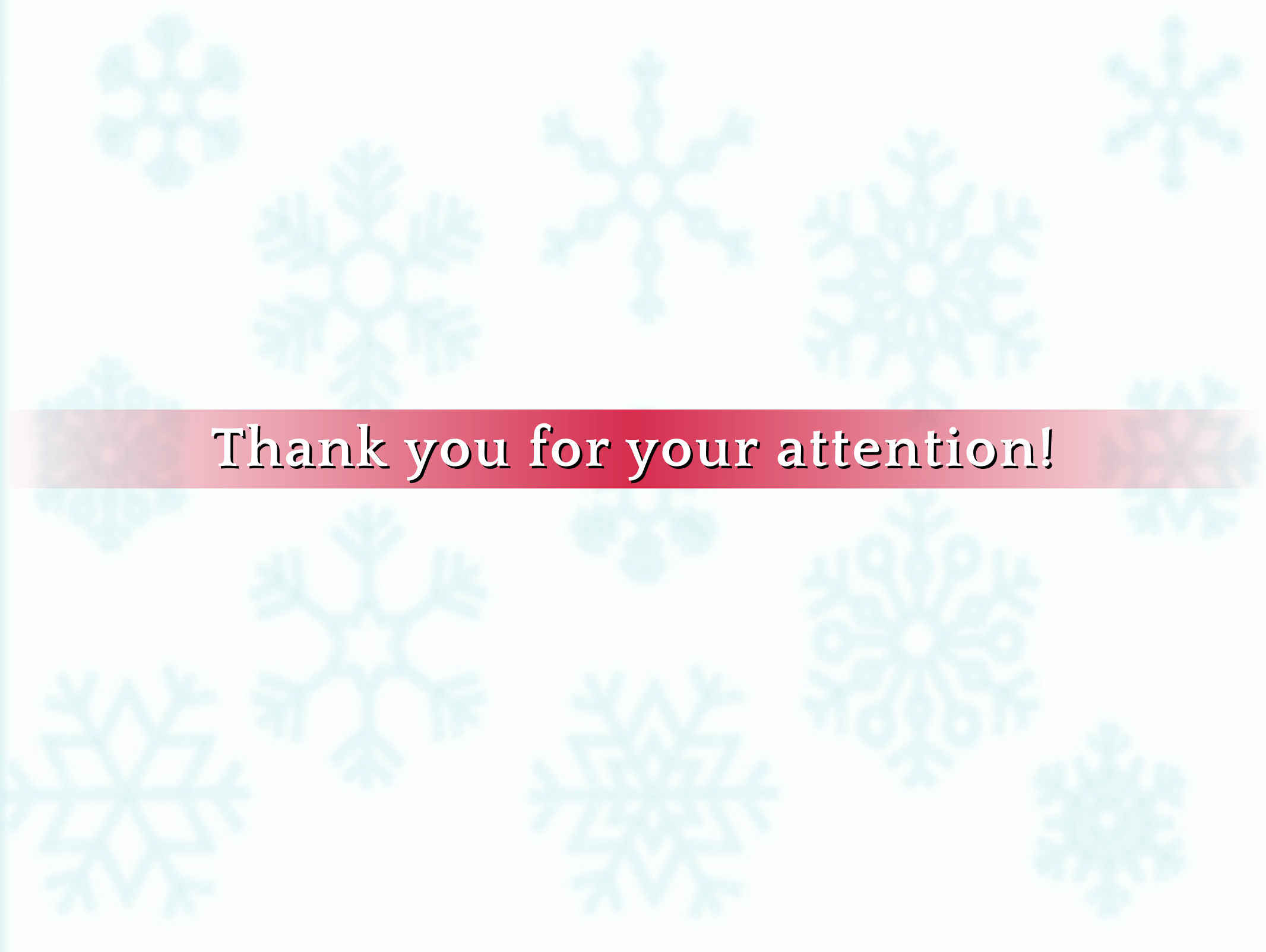
Transform an allowed perturbing matrix to Jordan-normal basis of defective matrix

↳ **its upper- ℓ Hessenberg structure reveals leading-order algebraic singularity $\sim \epsilon^{1/\ell}$**

Energy splitting $\propto \epsilon^{1/3}$ for PT

$\propto \epsilon^{1/2}$ for P

Different symmetries stabilize branch points of different orders



Thank you for your attention!